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COMPARATIVE ANALYSIS OF ELBOW JOINT LOADING IN PUSH-UP AND BENCH-PRESS

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ABSTRACT

Upper extremity pain and dysfunction are common orthopedic problems in individuals who practice physical activities involving the upper limbs. Push-ups (PUs) and bench-presses (BPs), representing closed kinetic chain (CKC) and open kinetic chain (OKC) exercises, respectively, are among the most popular training exercises for the upper extremity in sports training and clinical rehabilitation applications. Utilizing a self-developed exercise testing model, the present study examines the difference in elbow joint loading between OKC and CKC exercises of the upper extremity.

As many as 15 physically healthy male subjects with an average age of 19.8 years (S.D. 1.4), an average height of 176.8 cm (S.D. 4.2), and an average weight of 69.27 kg (S.D. 5.0) were studied. Each subject was asked to perform two different types of exercise, namely PUs and BPs. The kinematics, kinetics, and muscle activation of the elbow joint were calculated and analyzed using laboratory-developed motion analysis procedures.

The PU exercises result in a greater loading on the elbow joint than the BP exercises, but produce a lower muscle activation. OKC exercises yield a greater muscle strengthening effect than CKC exercises. The increased shear stress on the elbow joint should be carefully noted when performing PU exercises. The results presented in this study provide valuable information for treatment and rehabilitation purposes in clinical applications.

Keywords: Push-up; Bench-press; Elbow joint loading; Muscle activation; Rehabilitation.

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INTRODUCTION

Upper extremity pain and dysfunction are commonly observed in individuals who practice physical activities involving their upper limbs. In the search for effective rehabilitation programs, many previous studies have attempted to assess the variables (e.g. load and muscle activation/coactivation) in various exercises and motor tasks. Push-ups (PUs) and bench-presses (BPs) are among the most common training exercises for the upper extremity in sports training and clinical rehabilitation,[1] and can be classified as closed kinetic chain (CKC) and open kinetic chain (OKC) exercises, respectively.[2] In a previous study, Chou *et al.*[3] found that the elbow joint load in BP exercises reduces as the spacing between the hands is increased. The same group also examined the effect of the hand position on the intersegmental loading in PU exercises.[4] The results showed that PUs generate a significant valgus torque, and thus it was concluded that such exercises should be avoided in the immediate post-treatment rehabilitation of patients with medial collateral ligament damage or total elbow arthroplasty. However, while the variables in BP and PU exercises have been individually examined in previous studies, few attempts have been made to compare and contrast the elbow joint loading in CKC and OKC type exercises within a single study.

CKC and OKC exercises are often used for the rehabilitation and strengthening of the knee joint.[5–8] In the early 1990s, a shift away from OKC toward CKC exercises was observed for anterior cruciate ligament (ACL) rehabilitation since the most commonly used OKC knee extensor strengthening exercises were thought to increase the ACL strain values, whereas CKC exercises seemed to protect the graft from excessive strain while simultaneously restoring lower extremity muscle strength. However, Escamilla *et al.*[9] reported that the maximum anterior shear force acting on the knee in CKC exercises (0.67% body weight) was actually slightly higher than that in OKC exercises (0.44% body weight).

OKC and CKC exercises are commonly used in clinical rehabilitation and athletic training. The literature contains many investigations regarding the difference between OKC and CKC exercises of the lower extremity, and their clinical applications. However, the posture or loads in OKC and CKC exercises of the lower extremity (e.g. squats and knee extension exercises) vary from one form of exercise to another, and thus it is difficult to draw meaningful comparisons between the two sets of results. By contrast, the OKC and CKC exercises of the upper extremity conducted in the present study (i.e. PUs and BPs, respectively) both involve a similar load and posture, and thus the measured variables can be more reasonably compared.

During pitching, repeated stresses on the shoulder and elbow joints may cause pain and injuries to the overhead-throwing athletes. Before ball release, valgus torque around the elbow joint maybe as high as 64 N-m.[10] These repetitive stresses on the elbow could result in medial collateral ligament injuries (medial tension overload), osteochondritis dissecans of the capitellum (lateral compression overload), and impingement syndrome of the posterior compartment. This valgus extension overload was reported by Wilson *et al.*[11] as a common injury mechanism in overhead-throwing athletes. Modern strength training programs for athletes combine CKC and OKC exercises to improve core upper body and lower body strength, sport-specific strength, performance, and for injury prevention.[2] Even though PU and BP exercises are usually used for strengthening proximal stability, the elbow joint also plays an important role during the PU and BP exercises. These two exercises are also commonly used to strengthen the upper extremity and enhance sports performance.[2] During the two exercises, the largest range of motion occurred at the elbow joint. Therefore, understanding the biomechanical differences in the elbow joint loading between the two commonly used exercises, PU and BP, can provide valuable information for coaches and athletes in order to reduce the risk of sports injuries. Accordingly, this study utilizes a self-developed exercise testing model to examine the difference in elbow joint loading between OKC (PU) and CKC (BP) exercises of the upper extremity. The experimental results provide a useful source of reference for treatment and rehabilitation purposes in clinical applications.

METHODS

Subjects and Experimental Protocol

As many as 15 physically healthy male subjects with an average age of 19.8 years (S.D. = 1.4), an average height of 176.8 cm (S.D. = 4.2), and an average weight of 69.27 kg (S.D. = 5.0) took part in the PU and BP experiments. None of the subjects had any previous history of upper-extremity injuries or disorders.

During the experiments, the relative joint positions, ground reaction forces, and muscle activations were measured using an ExpertVison motion system (Motion Analysis Corp., Santa Rosa, CA, USA) comprising six 120 Hz CCD cameras, two 1000 Hz Kistler force plates (Type 9281B, Kistler Instrument Corp.,

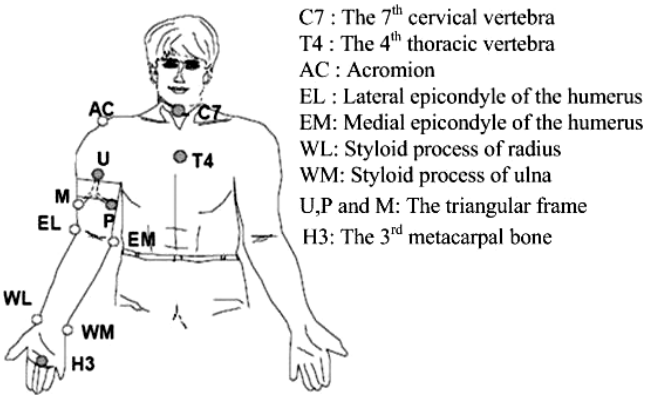


Fig. 1 Positioning of reflective markers on anatomical bony landmarks.

Winterthur, Switzerland), and the multi channel surface electromyographic (EMG) system (MA300, Motion Analysis Corp., USA).

A total of 11 reflective markers were placed on selected bony anatomic landmarks on each test subject. As shown in Fig. 1, eight of the landmarks were chosen in such a way as to simulate the rigid body assumption for the trunk (cervical vertebra 7, thoracic vertebra 4, and acromion), upper arm (acromion process, medial and lateral epicondyles of the elbow), forearm (medial and lateral epicondyles of the elbow, ulnar styloid processes), and hand (radial and ulnar styloid processes, third metacarpal bone), respectively, while three markers were attached to a triangular frame and placed on the upper arm in order to minimize the potential errors caused by movement of the epicondyles skin during the experiments. The shoulder joint center was defined as the point located at 90% of the length of an imaginary line drawn from the elbow joint center (as calculated by the medial and lateral markers) to the marker located at the acromion.[13]

The EMG sensors were attached to the pectoralis major, triceps-long head, biceps brachii, supraspinatus, deltoid anterior, deltoid middle, and deltoid posterior in order to measure the muscle activation during the PU and BP exercises. After placing the EMG electrodes and retro reflexive markers on the test subjects, the maximal voluntary contraction (MVC) was recorded for each muscle. The EMG activity of each muscle was normalized and presented in terms of %MVC. As shown in Fig. 2, the subjects were asked to perform two different types of exercises, namely PUs and BPs. During the two exercises, the subjects were asked to position their hands in such a way as to maintain a balanced state and to complete the exercises using a uniform motion. To achieve a similar loading condition in both the exercises, the BP exercises were performed using a 37-kg barbell, equivalent to the average loading condition in the "up" position in the PU exercises. The subjects were asked to perform the specified exercise continuously for a period of approximately 12 s. Each subject was then instructed to rest for a minimum of 30 min before commencing the next set of exercises in order to avoid muscle fatigue.

The exercise testing model was approved by both the IRB of Kaohsiung Medical University and the National Science Council of Taiwan. Moreover, consent-release agreements were signed by each subject prior to participation in the experiments. In addition, all of the experiments were carefully supervised in person by either the author or one of the coauthors (all with a sports medicine background) in order to minimize any potential risk to the volunteers.

Theorem and Governing Equations

The forces acting on the upper extremity joints during the PU and BP exercises were analyzed using a three-joint multi linkage system formed by the hand, forearm, upper arm, and trunk, respectively. The free body diagram of each joint (i.e. the wrist, elbow, and shoulder) is shown in Fig. 3, together with the corresponding joint forces and joint moments. The relevant governing equations are as follows.

From the free body diagram of the hand

$$\vec{F}_{\mathrm{hp}} = m_{\mathrm{h}}\vec{a}_{\mathrm{h}} - m_{\mathrm{h}}\vec{g} - \vec{F}_{\mathrm{hd}} \tag{1}$$

$$\vec{M}_{\mathrm{hp}} = I_{\mathrm{h}}\vec{\alpha}_{\mathrm{h}} - \vec{M}_{\mathrm{hd}} - (\vec{r}_{\mathrm{hp}} \times \vec{F}_{\mathrm{hp}}) - (\vec{r}_{\mathrm{hd}} \times \vec{F}_{\mathrm{hd}}) + \vec{\omega}_{\mathrm{h}} \times (I_{\mathrm{h}} \cdot \vec{\omega}_{\mathrm{h}}) \tag{2}$$

from the free body diagram of the forearm

$$\vec{F}_{\mathrm{fd}} = -\vec{F}_{\mathrm{hp}} \tag{3}$$

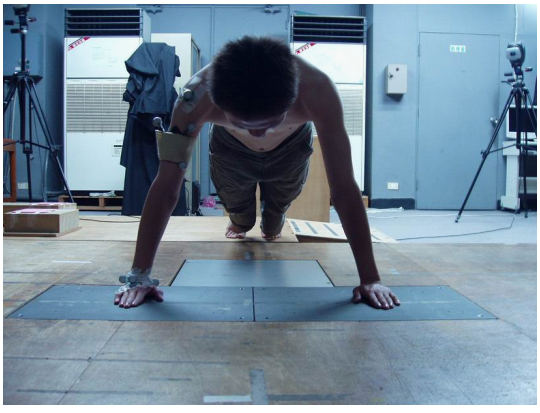
$$\vec{M}_{\mathrm{fd}} = -\vec{M}_{\mathrm{hp}} \tag{4}$$

$$\vec{F}_{\mathrm{fp}} = m_{\mathrm{f}}\vec{a}_{\mathrm{f}} - m_{\mathrm{f}}\vec{g} - \vec{F}_{\mathrm{fd}} \tag{5}$$

$$\vec{M}_{\mathrm{fp}} = I_{\mathrm{f}}\vec{\alpha}_{\mathrm{f}} - \vec{M}_{\mathrm{fd}} - (\vec{r}_{\mathrm{fp}} \times \vec{F}_{\mathrm{fp}}) - (\vec{r}_{\mathrm{fd}} \times \vec{F}_{\mathrm{fd}}) + \vec{\omega}_{\mathrm{f}} \times (I_{\mathrm{f}} \cdot \vec{\omega}_{\mathrm{f}}) \tag{6}$$

Note that the notations in the equations above are defined as follows

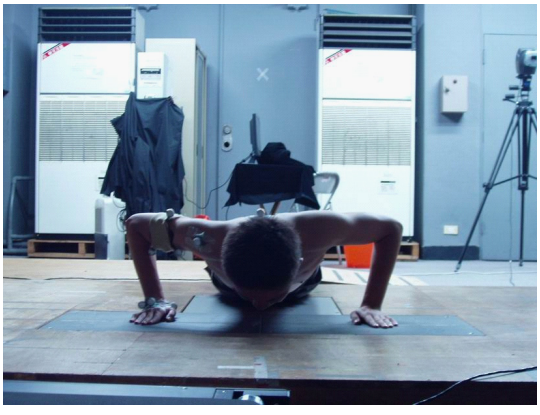
h = Hand, f = Forearm,
$\vec{F}_{\mathrm{p}}$: Proximal joint force,
$m\vec{a}$: Effective force,
$m\vec{g}$: Gravity force acting on local segment,
$\vec{F}_{\mathrm{d}}$: Distal joint force,
m : Mass of segment,
$\vec{M}_{\mathrm{p}}$: Proximal joint moment,



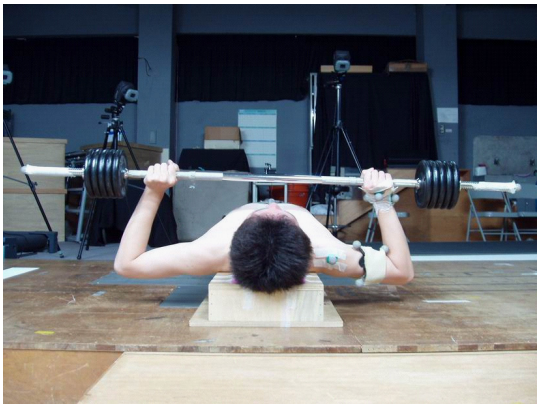
(A) The push-up exercise at"up" position.



(B) The bench-press exercise at "up" position.



(C) The push-up exercise at"down"position.



(D) The bench-press exercise at"down" position.

Fig. 2 Photographs of push-up and bench-press exercises.

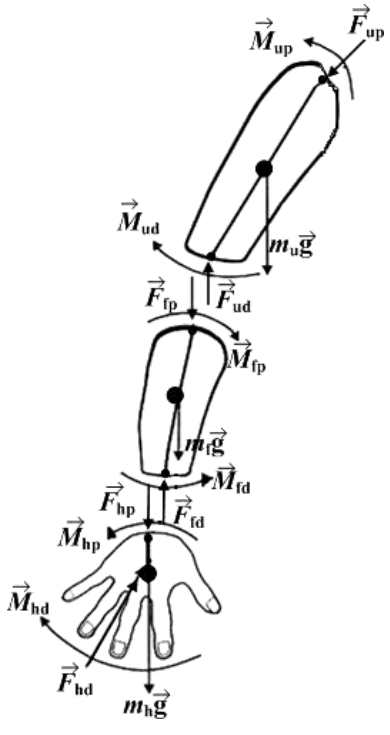


Fig. 3 Segmental free body diagram of upper extremity.

I : Mass moment of inertia,

$\vec{\alpha}$: Angular acceleration of local segment,

$\vec{M}_{\mathrm{d}}$: Distal joint moment,

$\vec{r}_{\mathrm{p}}$: Rotation matrix describing relative rotation between local coordinates of proximal segment and global coordinates,

$\vec{r}_{\mathrm{d}}$: Rotation matrix describing relative rotation between local coordinates of distal segment and global coordinates,

$\vec{\omega}$: Angular velocity of local segment.

The kinematics and kinetics data were obtained experimentally. In accordance with Newton's third law, the force and moment acting on any given joint were assigned opposite signs when substituted into the governing equations of the proximal body segment.

Data Reduction

The joint angles, resultant forces, and moments acting on the elbow were calculated using laboratory-developed kinematics and kinetics software.[14–16]

The kinematics and kinetics analyses were based on a three-segment model comprising the hand, forearm, and upper arm. Note that each segment was assumed to be a rigid body. Three elbow joint angles (i.e. the hinge angle, rotational angle, and horizontal deviation) were then calculated using Euler's method with a $y - x - z$ rotational sequence.[17,18] In performing the analysis, the elbow flexion angle was defined as the range of motion (ROM) of elbow flexion during the related exercise. In the PU exercises, a piezoelectric force plate was used to measure the vertical and shear forces. In the BP exercises, a constant loading 37 kg barbell was used as external load to hand. For both PU and BP exercises, simultaneous measurements of the upper-extremity kinematics were obtained by capturing the displacements of the reflective markers. The segment mass and inertial data were estimated by anthropometry,[19] while the angular velocity and acceleration data were calculated using Euler's parametric method.[14] The force plate loading was assumed to be equal and opposite to the hand loading. The wrist loading and wrist moment were calculated using an inverse dynamic procedure based on the Newton–Euler equations.[14–16] Finally, the elbow joint loading was calculated using the governing equations given in the section on "Theorem and Governing Equations".

Data smoothing was accomplished using a generalized cross validation spline (GCVSPL) smoothing routine with a cut-off frequency of 6 Hz.[20] The joint angles and joint forces acting on the elbow were calculated as a function of time over the course of the exercise period and were then used for analysis purposes.[14,16,21,22]

The raw EMG data were filtered using a 60 Hz notch filter with a frequency bandwidth of 10 to 500 Hz. For each trial, the EMG data were normalized as a percentage of the corresponding MVC. The normalized data were then used for statistical analysis purposes.

Data Analysis

The difference in the ground reaction force, joint angle, joint force, and muscle activation between the PU and BP exercises was evaluated by means of the ANOVA statistical analysis technique using SPSS software with a statistical significance level of 5%. Captured data at the up, down, and peak events of an entire PU and BP press were used in the following analysis. The "up" event was defined as the moment of initial posture, with the elbow in full extension. The "down" event was defined as the moment of the chest being at its lowest position. The "peak" event was defined as the moment when the kinematics or kinetic data was either maximal or minimal. In performing the analysis, it was hypothesized that there was no significant difference in the elbow loading, elbow flexion angle, or muscle activation between the PU and BP exercises, respectively.

RESULTS

Elbow Flexion Angle

The peak flexion angle of the elbow joint during the PU and BP exercises was 93 degrees in both cases (Fig. 4). In other words, there was no significant difference between the elbow flexion angle in the CKC and OKC exercises, respectively.

Elbow Joint Force

The mediolateral (ML) shear forces in the PU exercises were significantly greater than those in the BP exercises (Table 1). However, the anteroposterior (AP) shear forces in the PU exercises were not significantly different than those in the BP exercises. The axial compressive force was the dominant force in both types of exercise. The peak axial compressive forces in the PU and BP exercises were 40.7% and 27.6% body weight, respectively. The axial compressive forces in the PU experiments were significantly larger than those in the BP experiments.

Elbow Joint Moment

The valgus moments of the elbow in the BP experiments were significantly lower than those in the PU group ($P < 0.05$, Table 2). The peak valgus moment in the PU experiments averaged 13.86 Nm, while that in the BP experiments averaged 5.23 Nm.

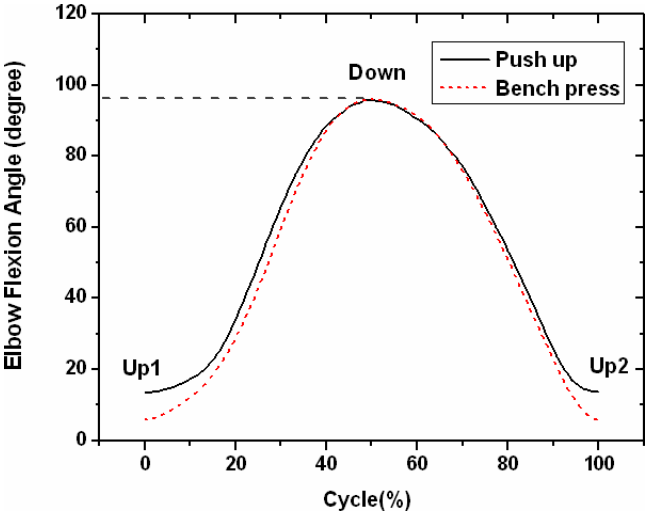


Fig. 4 The elbow flexion angle during push-up and bench press exercise.

Table 1. Joint Forces Acting on Elbow During Push-Up and Bench-Press Exercises.

The Joint Force of Elbow During Open/ Closed Chain Exercises			
	Push-Up Mean (SD)	**Bench-Press Mean (SD)**	**P**
Medial(+)/Lateral(−) force			
Up	4.5(2.5)	0.8(2.2)	*
Down	8.4(4.4)	7.3(3.3)	0.41
Peak	12.3(3.9)	9.2(2.1)	*
Anterior(+)/Posterior(−) force			
Up	−2.8(2.7)	−3.4(1.9)	0.38
Down	−4.8(3.0)	−2.9(4.6)	0.20
Peak	−6.2(3.0)	−3.5(2.1)	0.15
Axial Compression(+) force			
Up	28.1(3.5)	27.3(2.7)	0.59
Down	38.5(2.5)	26.4(2.9)	*
Peak	40.7(3.0)	27.6(2.7)	*

Note: # P value is significance of one-way ANOVA; "*" P < 0.05; unit % body weight.

Table 2. Joint Moments Acting on Elbow During Push-Up and Bench-Press Exercises.

The Joint Moment of Elbow During Open/Closed Chain Exercises			
	Push-Up mean (SD)	**Bench-Press mean (SD)**	**P**
Varus(+)/Valgus(−)			
Up	−1.43(6.6)	1.12 (2.6)	0.48
Down	−10.62(6.9)	4.68 (9.7)	*
Peak	−13.86(8.1)	5.23 (9.0)	*
Flexion(+)/Extension(−)			
Up	−11.42(5.2)	−3.68(4.4)	*
Down	−15.38(9.3)	−14.28(6.5)	0.48
Peak	−23.77(9.9)	−20.85(4.9)	0.08
Supination(+)/Pronation(−)			
Up	−0.10(1.0)	−1.35(0.7)	*
Down	−2.01(1.7)	1.81 (1.5)	0.73
Peak	−2.85(1.7)	2.32 (0.8)	0.14

Note: # P value is significance of one-way ANOVA; "*" $P < 0.05$; unit % body weight.

Muscle Activation

The peak muscle activation appeared in the pectoralis major in both the PU and the BP exercises, and had a value of 86% MVC and 100% MVC, respectively (Fig. 5). The peak muscle activations in the pectoralis major, triceps-long head, biceps brachii, supraspinatus, deltoid anterior, and deltoid middle in the BP experiments were significantly greater than those in the PU experiments.

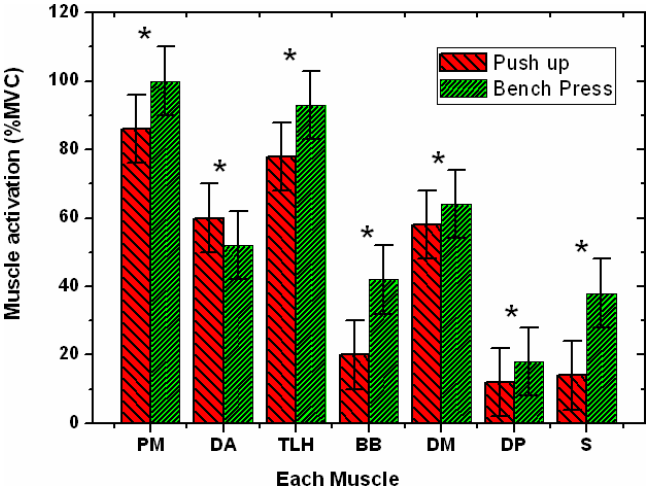


Fig. 5 The normalized muscle activation during push-up and bench-press exercise.

Bars represent SD. ("*": P < 0.05)
Unit: %MVC
Muscle activation = $\frac{\text{The EMG value during an exercise cycle}}{MVC}$
MVC: the maximal isometric voluntary contraction
The definition of each muscle:
PM: Pectoralis Major, **DA:** Deltoid Anterior, **TLH:** Triceps-Long Head, **BB:** Biceps Brachii, **DM:** Deltoid Middle, **DP:** Deltoid Posterior, **S:** Supraspinatus.

DISCUSSION

Various studies have shown that OKC knee extensor strengthening exercises increase the ACL strain value.[4,5,7] Accordingly, CKC exercises have been proposed as a more functional exercise capable of both protecting the graft from excessive strain and simultaneously restoring lower extremity muscle strength in the early stage after the ACL reconstruction.[6]

In lower extremity exercises, the shear force acting on the joint depends on the loading direction relative to the joint, which varies in turn with the particular exercise. By contrast, in the CKC and OKC upper extremity exercises performed in the present study, it appears that the same loading direction and movement yield notable differences in the force acting on the elbow joint. In this study, the axial compression force, anterior/posterior force, and medial/lateral shear force in the PU exercises (CKC) were greater than in the BP exercises (OKC). In other words, the OKC exercise results in a lower strain on the elbow.

Anderson and Behm[23] investigated the EMG activity in different muscle groups during PU, BP, and wall-press exercises performed with the hands supported on a table or a relatively unstable surface. The results indicated that the greatest mean electrical activation amplitude occurred in the pectoralis major during the PU exercises. By contrast, in the present study, the

greatest muscle activation occurred in the pectoralis major during the BP exercises. In practice, the difference in findings of the two studies regarding the exercise with the greatest electrical activation amplitude most likely reflects the difference in the external force and body movement characteristics of the two sets of results. For example, in the present study, the external force in the two types of exercise is similar (i.e. 37 kg), whereas in the investigation of Anderson and Behm,[23] the external force varied from one type of experiment to the next. However, in performing BP exercises, the subjects cannot generally sustain the same load as that generated in the "up" position of the PU exercises. Thus, the subjects will invariably perform the BP exercises at a lower rate than the PU exercises.

During both PU and BP exercise, the cycle can be divided to two strokes: (1) descending movement (up to down), and (2) ascending movement (down to up). The upper extremity consists of the hand, forearm, and upper arm. At the beginning of both PU and BP exercises, the upper extremity is at the "up" position. At this position, the upper extremity works as a column to bear weight. From the viewpoint of the mechanics, the upper extremity becomes the best regular structure for the weight bearing. When the elbow joint performed flexion motion, the best regular structure for weight bearing will be broken. Especially, the regular structure will become the worse singular structure for weight bearing at certain phase between the elbow angles at 90° to "down," during the descending process. Similarly, this phenomena also occurs at the elbow angles between "down" to 90° during the ascending process. This is why the significant peak of elbow joint force occurs at this particular phase.

During pitching, repeated stresses on the shoulder and elbow joints may cause pain and injuries to the overhead-throwing athletes. Before ball release, valgus torque around the elbow joint maybe as high as 64 N-m.[10] These repetitive stresses on the elbow could result in medial collateral ligament injuries (medial tension overload), osteochondritis dissecans of the capitellum (lateral compression overload), and impingement syndrome of the posterior compartment. This valgus extension overload was reported by Wilson *et al.*[11] as a common injury mechanism in overhead throwing athletes. Modern strength training programs for athletes combine CKC and OKC exercises to improve core upper- and lower body strength, sport-specific strength, performance, and for injury prevention.[2] Therefore, understanding the biomechanical differences between the two commonly used exercises, PU and BP, can provide valuable information for coaches and athletes in order to reduce the risk of sports injuries.

Comparing the PU and BP exercises, it is interesting to note that the joint loads in the PU experiments are greater than those in the BP experiments. The peak medio-lateral loading of the elbow joint was decreased by 25.2% in the BP exercise. Additionally, the peak muscle activations of triceps-long head and biceps brachii, which are major contributor of the elbow joint, are also significantly greater in the BP experiments. These findings indicated that OKC exercise (BP) can be used as an effective strengthening program for the upper extremity.

CONCLUSIONS

This study has shown that significant differences exist in the elbow joint loading produced by CKC and OKC exercises of the upper extremity (i.e. PUs and BPs, respectively). The axial compressive force represents the dominant force in both the PU and BP exercises. The axial compressive force has a value of 40.6% body weight in the PU experiments and 27.6% body weight in the BP experiments. The axial compressive forces in the "down" and "peak" events in the PU exercises are significantly larger than those in the BP experiments. In addition, the peak valgus moment of the elbow in the PU experiments has an average value of 13.86 Nm, whereas that in the BP experiments has an average value of 5.23 Nm.

The peak muscle activation occurs in the pectoralis major in the OKC exercises and is around 14% MVC higher than the equivalent value in the CKC exercises. Furthermore, the peak muscle activations of the pectoralis major, triceps-long head, biceps brachii, supraspinatus, deltoid anterior, and deltoid middle in the BP experiments are significantly higher than those in the PU experiments.

Overall, the results in this study show that OKC (BP) exercises of the upper extremity produce a greater muscle activation and a lower elbow joint loading than CKC (PU) exercises. Therefore, BP exercises represent an effective strengthening program for the upper extremity. The experimental data presented in this study provides valuable information for the treatment and rehabilitation of the upper extremity in clinical applications.

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